Shihan Kanungo

Palo Alto, California, United States of America
Personal website | shihankanungo@gmail.com | (650)643-7971

San José State University

Concurrent enrollment during high school. Graduate courses indicated by †

A+ **MATH 279A † Graph Theory.** Yan Zhang — Spring 2024
Douglas B. West. *Introduction to Graph Theory.* Prentice Hall 1996

A **MATH 221A † Higher Algebra I.** Edgar Bering — Fall 2025
Paulo Aluffi. *Algebra: Chapter 0.* AMS 2021. Covered Chs. 1–4.

A+ **MATH 221B † Higher Algebra II.** Edgar Bering — Spring 2026
Paulo Aluffi. *Algebra: Chapter 0.* AMS 2021. Covered Chs. 5–9.

A+ **MATH 226 † Theory of Numbers.** Jordan Schettler — Spring 2024
Primes in arithmetic progressions, partitions, modular group, and the Dedekind eta function.

A+ **MATH 285M † Elliptic Curves and Modular Forms.** Jordan Schettler — Fall 2025
Alvaro Lozano-Robledo. *Elliptic Curves, Modular Forms, and Their L-functions.* AMS 2011.

A+ **MATH 229 † Advanced Matrix Theory.** Wasin So — Fall 2024
R. Horn & C. Johnson. *Matrix Analysis*, Cambridge University Press 2012.

A+ **MATH 285M † Nonnegative matrices.** Wasin So — Spring 2025
The basic theory of nonnegative matrices; applications to graphs and digraphs.

A+ **MATH 275A † Topology.** Yan Zhang — Fall 2025
Compactness, connectedness; Tychonoff's Theorem; paracompactness and metrization theorems.

A+ **MATH 231A † Real Analysis.** Slobodan Simic — Fall 2024
E.M.Stein & R.Shakarchi, *Real Analysis: Measure Theory, Integration, and Hilbert Spaces*, Chs. 1–4.

A+ **MATH 231B † Functional Analysis.** Slobodan Simic — Spring 2025
L Debnath & P Mikusinski. *Introduction to Hilbert Spaces with Applications.* Academic Press 2005.

A+ **MATH 238 † Advanced Complex Variables.** Jordan Schettler — Spring 2026
E.M.Stein & R.Shakarchi, *Complex Analysis*, Chs. 1–5, 8.

A+ **MATH 233A † Partial Differential Equations.** Daniel Brinkman — Spring 2025
Peter J Olver. *Introduction to Partial Differential Equations.* Springer 2013.

A+ **MATH 234 † Advanced Dynamical Systems.** Liam Stanton — Spring 2024
Continuous and discrete systems; stability of equilibria and closed orbits, structural stability.

A **MATH 285A † Stochastic Methods in Scientific Computing.** Liam Stanton — Spring 2025
Monte Carlo methods, Metropolis-Hastings algorithm, random-number generation.

A **MATH 129A Linear Algebra I.** Dashiell Fryer — Fall 2023
S.H.Friedberg, et al. *Linear Algebra* 2nd ed., Prentice Hall 1989.

A+ **MATH 129B Linear Algebra II.** Wasin So — Spring 2024
S.H.Friedberg, et al. *Linear Algebra* 2nd ed., Prentice Hall 1989.

A+ **MATH 131A Introduction to Analysis I.** Tim Hsu — Spring 2024
Kenneth Ross, *Elementary Analysis: The Theory of Calculus.* UTM, Springer NY 2013.

A+ **MATH 131B Introduction to Analysis II.** Tim Hsu — Fall 2024
Tim Hsu, *Fourier Series, Fourier Transforms, and Function Spaces.* AMS/MAA 2020

A+ **MATH 138 Complex Variables.** Liam Stanton — Fall 2023
Dennis G. Zill et al, *A First Course in Complex Analysis with Applications.* Jones & Bartlett 2006.

A+ **MATH 179 Introduction to Graph Theory.** Sogol Jahanbekan — Fall 2024
R. J. Wilson. *Introduction to Graph Theory.* Prentice Hall 2010

Euler Circle

College-level math classes taught by Simon Rubinstein-Salzedo. Grades not assigned.

Abstract Algebra. Spring 2020
Group theory, Sylow theorems, fields and extensions, number fields, Galois correspondence.

Ring Theory and Algebraic Geometry . Summer 2026
Affine and projective algebraic geometry, Hilbert's Nullstellensatz, localization and local rings, affine schemes and morphisms.

Combinatorics. Fall 2025
Special numbers; sieving; restricted permutations; posets; incidence algebras; generating functions.

Combinatorial Game Theory. Fall 2024
Nim, Hackenbush, surreal numbers, impartial games, Sprague-Grundy theory.

Generating Functions. Fall 2021
Philippe Flajolet and Robert Sedgewick. *Analytic Combinatorics.* Cambridge University Press 2011.

Number Theory. Fall 2023
Reciprocity theorems, quadratic forms, elliptic curves, modular curves.

Analytic Number Theory. Spring 2024
Dirichlet's Theorem, The Prime Number Theorem, Brun's theorem, smooth numbers.

Complex Analysis I. Spring 2026
Cauchy integral formula, residues & series, branch points, Riemann surfaces, products, Gamma function.

Complex Analysis II. Summer 2026
Hadamard product, Riemann mapping thm., Picard's little thm., Wiener-Ikehara thm., prime number thm., Theta functions.

Ergodic Theory. Winter 2024
Simon Rubinstein-Salzedo. *Ergodic Theory*, The Carus Mathematical Monographs. MAA Press 2025.

Differential Geometry. Spring 2022
Andrew Pressley. *Elementary Differential Geometry.* Springer, 2001.

Markov Chains. Fall 2020
D.A. Levin et al, *Markov chains and mixing times.* American Mathematical Society 2006

Representation Theory

I haven't taken a formal course, but have studied the subject through the following books.

Introduction to Lie algebras. 2006
Karin Erdmann. Springer Verlag, London.

Introduction to Lie Algebras and Representation Theory. 1972
Humphreys, J.E. Springer Verlag, New York.

Representations of Semisimple Lie Algebras in the BGG Category $\mathcal{O}$. 2008
Humphreys, J.E. Graduate Studies in Mathematics **94**, AMS.

Lie Superalgebras and Enveloping Algebras. 2012
Ian. M. Musson. Graduate Studies in Mathematics **131**, AMS.

Yangians and Classical Lie algebras. 2007
Alexander Molev. Mathematical Surveys and Monographs **143**, AMS.

Dualities and Representations of Lie Superalgebras. 2012
Cheng, S. and Wang, W. Graduate Studies in Mathematics **144**, AMS.

Lectures on Symmetric Tensor Categories. 2021
Pavel Etingof and Arun Kannan. [arXiv]